

Optimization-Based Decision Support for Equitable FQHC Expansion in Chicago

Meeting with Prof. Sage Kim

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April 19, 2026

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3. **The MUA designation process is uneven.** 87 census tracts in Chicago meet IMU eligibility but were never designated, predominantly in Black South Side neighborhoods (Kim et al., 2020).

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3. **The MUA designation process is uneven.** 87 census tracts in Chicago meet IMU eligibility but were never designated, predominantly in Black South Side neighborhoods (Kim et al., 2020).
4. **No referral network analysis.** HRSA mandates referral relationships with hospitals and specialists, but no one has mapped how referrals actually flow in Chicago.

The Optimization Model in Plain English

Inputs: facility locations, demand, capacity, travel constraints.

Output: many high-quality configurations for decision-makers to *compare and choose from* — not a single optimal solution.

Two hierarchical levels:

- **Districts** — catchment areas around each primary care facility. Addresses HRSA's catchment area design requirement.
- **Superdistricts** — groups of primary care districts sharing a referral hospital. Addresses HRSA's referral relationship requirement.

The framework is built to respect equity, travel time by public transit, and user-defined need measures.

Near-Term Paper: FQHC Optimization with UDS Projection

Data gap. UDS data (visits, FTEs) is reported at the *organization* level, not per site. 25 parent organizations operate 166 FQHC sites in Chicago.

Our approach. Project aggregate organization-level UDS onto individual sites proportionally — by provider count, facility size, or reported FTEs.

What this enables.

- Run the full FQHC optimization framework with realistic demand and capacity estimates.
- Publish the methodology now, without waiting on a data use agreement.
- Demonstrate the analytical value of having accurate site-level data.

Honest caveat. This is an approximation. Its purpose is to motivate the case for true site-level data.

Follow-up: Chicago Physician Referral Network

Data source. CMS Physician Shared Patient Patterns — physicians are nodes, shared Medicare patients within 30-day windows form edges.

What we would build.

- A Chicago-specific physician referral network.
- Overlay with FQHC and hospital data to identify *referral deserts* — communities where patients have no viable path from primary care to specialist care.

Connection to your CBO capacity work. Referral pathways combined with real-time capacity data is what turns a static map into a dynamic routing tool.

Data access. ResDAC application, or a data use agreement through UIC.

What We Offer and What We Need

What we bring:

- A spatial optimization framework for hierarchical facility location and districting.
- Mathematical methods for equitable resource allocation.
- An interactive decision-support dashboard integrating 166 FQHC sites, 128 hospitals, 61,460 providers, and 33 health indicators.

What we need:

- Health domain expertise to ground the modeling choices.
- Data access through UIC (site-level UDS, CMS referral data).
- Understanding of what decision-makers actually need from such tools.